

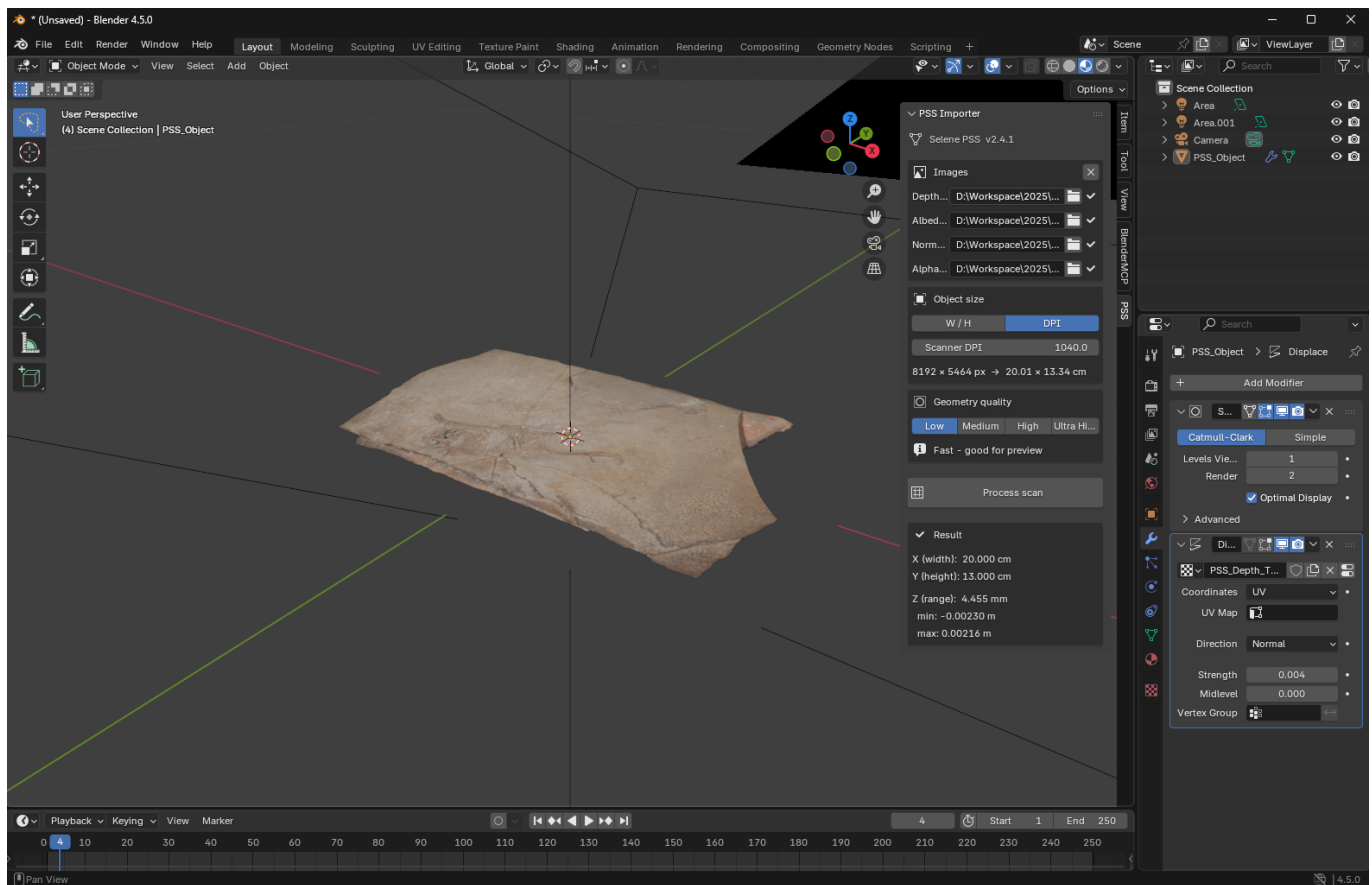
Selene PSS Importer — Blender addon

User manual v2.4

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What this addon does

The Selene PSS addon imports photometric stereo datasets into Blender, building a displaced 3D mesh from a 32-bit float depth map and applying albedo, normal map, and alpha textures.



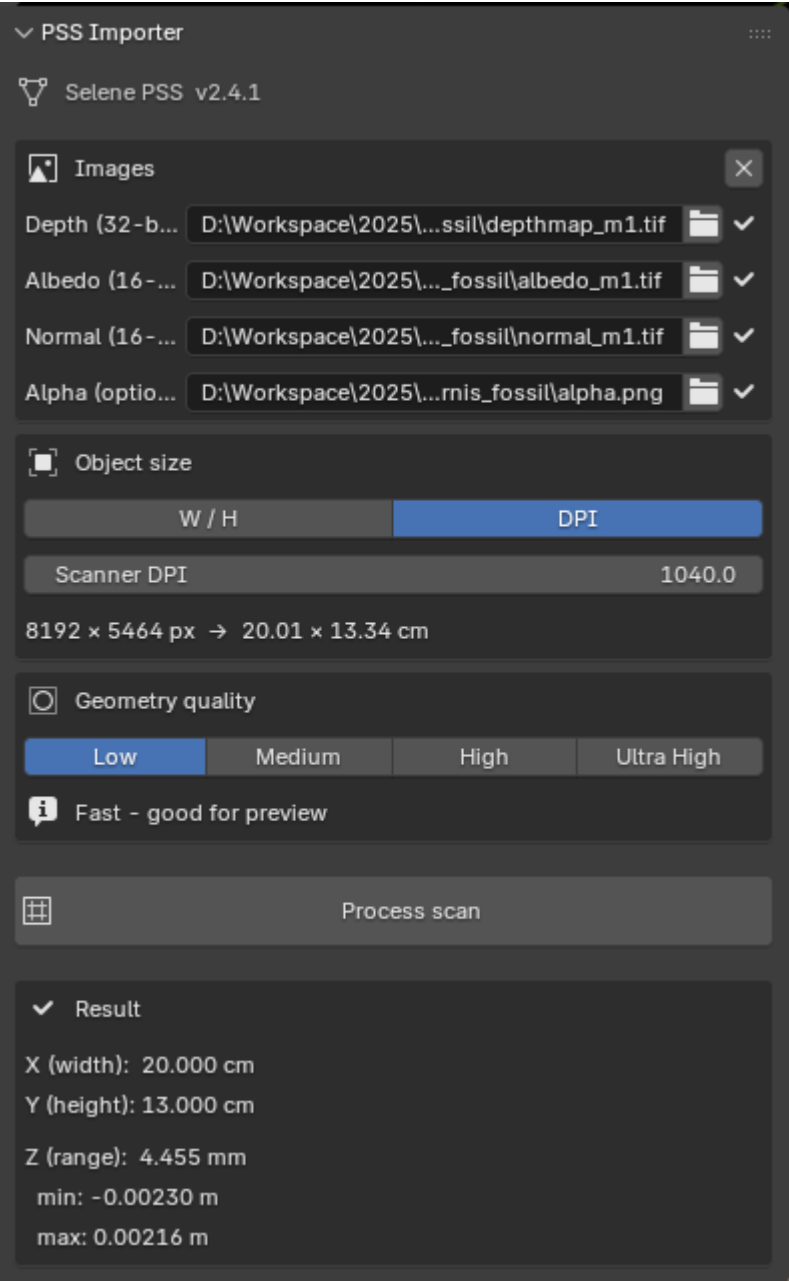
Requirements

- **Blender 4.1 or newer** (drag-and-drop requires 4.1+)
- **Depth map:** 32-bit float grayscale TIFF with values in metres (as exported by Selene PSS)

Installation

1. Download [pss_importer_addon.py](#)
2. In Blender: **Edit** → **Preferences** → **Add-ons** → **Install**
3. Select [pss_importer_addon.py](#) and click **Install Add-on**
4. Enable **Selene PSS** in the list
5. Open the **N-panel** in the 3D viewport (press **N**) and select the **PSS** tab

The interface



The panel is divided into four sections:

Images

Four slots for input files:

Slot	Format	Required
Depth (32-bit)	32-bit float TIFF — depth map in metres	Yes
Albedo (16-bit)	16-bit RGB TIFF	No
Normal (16-bit)	16-bit RGB TIFF	No
Alpha (optional)	8-bit PNG	No

Each slot has a **folder button** to browse for files. A checkmark icon appears when a valid file is assigned.

The **X button** (top right of the Images box) clears all slots at once.

Drag and drop (Blender 4.1+)

Files can be assigned by **dragging and dropping directly onto the 3D viewport**. The addon detects the slot automatically based on the filename:

Keywords in filename	Assigned to
depth, depthmap, disp	Depth
albedo, color, colour, rgb	Albedo
normal, nrm	Normal
alpha, mask	Alpha

If no keyword matches, the file goes to the first empty slot in order.

Object size

Two modes are available:

W / H mode — enter the real-world dimensions of the scanned object directly in centimetres.

DPI mode — enter the scanner resolution in DPI. The addon calculates the physical size automatically:

```
width_cm  = image_width_px  / dpi × 2.54
height_cm = image_height_px / dpi × 2.54
```

For Selene PSS scans, check the metadata **.txt** file for resolution, or use the values from the scan configuration.

Geometry quality

Controls the subdivision density of the base mesh:

Setting	Subdivision factor	Use
Low	n = 64	Quick preview
Medium	n = 32	General use
High	n = 16	Final quality
Ultra	n = 8	Maximum detail (slow)

Lower n = more subdivisions = more geometric detail but higher memory and processing time.

Processing a scan

1. Assign the depth map (required) and any additional maps
2. Set the object size (W/H or DPI)
3. Choose geometry quality
4. Click **Process scan**

The addon will:

- Load the 32-bit depth map and extract the real depth range in metres
- Apply linear min-max stretch and save a 16-bit displacement TIFF alongside the source file (*_disp16b.tif)
- Build a subdivided grid plane with the correct real-world dimensions
- Apply a Displace modifier with `strength = depth_range_m` and `mid_level = 0.0`
- Assign albedo, normal map, and alpha textures to a Principled BSDF material
- Set up basic lighting and camera

Results

After a successful process, the **Result** box shows:

- **X / Y**: actual plane dimensions in cm
- **Z range**: total displacement range in mm
- **Z min / max**: raw depth values from the TIFF in metres

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Viewing and navigating

Viewport controls

Action	Control
Rotate	Middle mouse button
Pan	Shift + Middle mouse
Zoom	Scroll wheel
Frame object	Numpad .

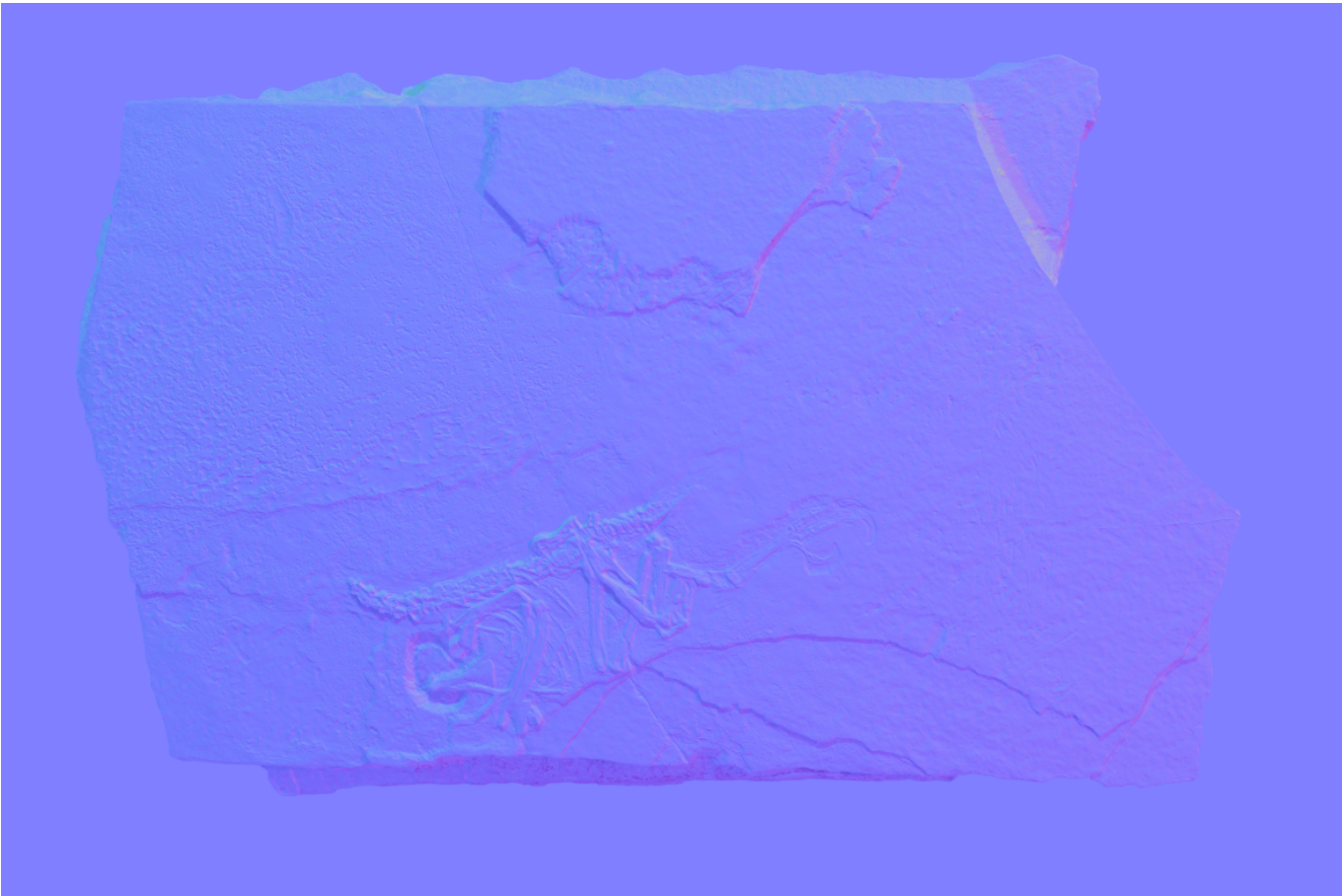
Shading modes

In the top-right of the 3D viewport (four sphere icons):

- **Material Preview** — shows textures (active after import)
- **Rendered** — full quality with lighting

About the normal map

The normal map encodes surface orientation at pixel level, adding fine detail without extra geometry. In the Selene PSS pipeline it is computed directly from the photometric stereo captures.



Exporting

For Sketchfab or web viewers (glTF)

- 1. Select the object
- 2. **File** → **Export** → **glTF 2.0 (.glb/.gltf)**
- 3. Settings:
 - Format: **glTF Binary (.glb)**
 - Include: **Selected Objects**
 - Mesh: **Apply Modifiers** (required to bake displacement)

For 3D printing (STL)

- 1. Select the object
- 2. **File** → **Export** → **STL (.stl)**

Troubleshooting

Symptom	Likely cause	Fix
Model is completely flat	Depth map not read correctly	Check that the TIFF is 32-bit float; check console for [INFO] Depth range
Wrong physical scale	DPI value incorrect	Switch to W/H mode and enter dimensions manually
Drag-and-drop assigns to wrong slot	Filename has no recognisable keyword	Rename file to include depth , albedo , normal , or alpha

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